



Documentation

CHROMA - Laser Arena



Thank you!

Thank you for choosing this pack! We hope you create something really special with it.

Please consider rating the package through your download list or leaving a review on the store page once you're familiar with it.

Feel free to give us feedback via E-Mail info@tidalflask.com or on our social media!

Your feedback helps us focus on the right updates for the future, which will be free for existing users!

*Enjoy, your **Tidal Flask** team!*





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Quick Start

Importing to Built-in RP project

After importing the Standard version into your Unity project 2021.3.20 & above, which doesn't use any of the Scriptable render pipeline packages (URP/HDRP), it should just worktm.

If you see any warnings in the Console window, try the Clear button and/or relaunch Unity. If the warnings don't disappear consult the FAQ or drop us an e-mail.

If you see any pink assets inside the Project window or in the scenes, simply select said asset -> right click -> Reimport and it should fix it. If you still encounter pink shaders, please make sure you have the correct pack version installed and that you are using a Unity version that is compatible with the pack.

Make sure you have Post Processing installed from Unity's Package Manager. If you install it after you imported the pack, reload the demoscene to get rid of possible errors.

Importing to URP project

Additionally to the built-in RP version, this pack also includes a version which works with the Universal Render Pipeline. If you want to find out exactly what it can and can't do please visit this page:

<https://docs.unity3d.com/Manual/render-pipelines.html>

Make sure you are importing the URP version of our package if you are using URP and Unity version 2021.3.20 or above.

On the following pages you will find detailed steps on how to import the package.

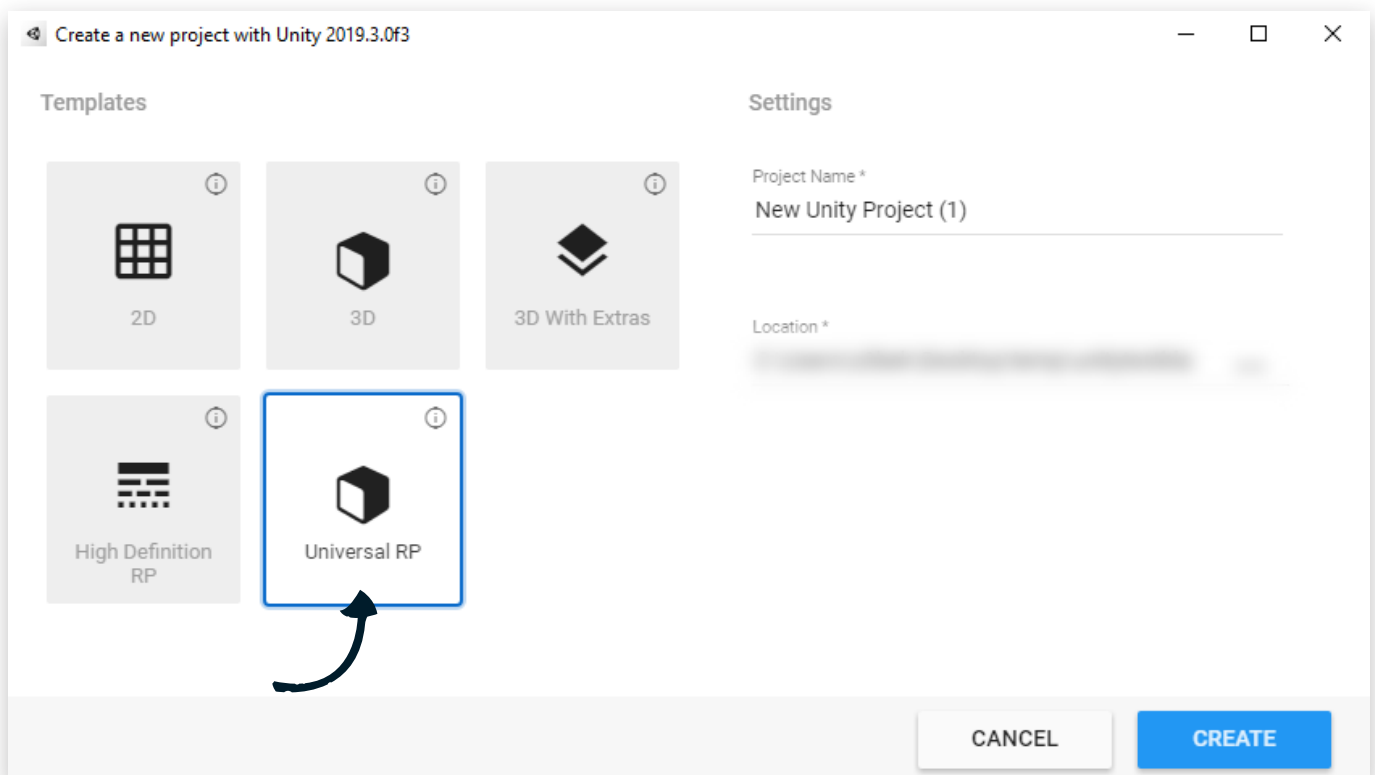


How to set up your project for URP (option 1)

We recommend to create a clean project and install the URP via the Package Manager or via Templates and import our package to this project.
To do so follow the steps below:



Step 1: Click “NEW” to create a new project (for URP pick Unity 2021.3.20 or above).



Step 2: In the “Templates” select “Universal RP”, this way everything you need for this package will be preinstalled.

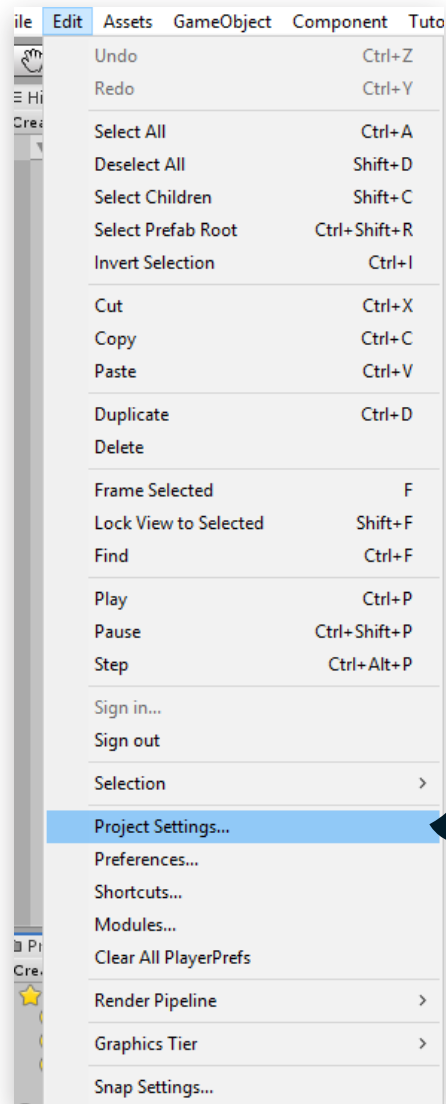


Step 3: Download the pack from the Asset Store and install the URP version.
At this point you already can go to the scenes folder and select any of the scenes.

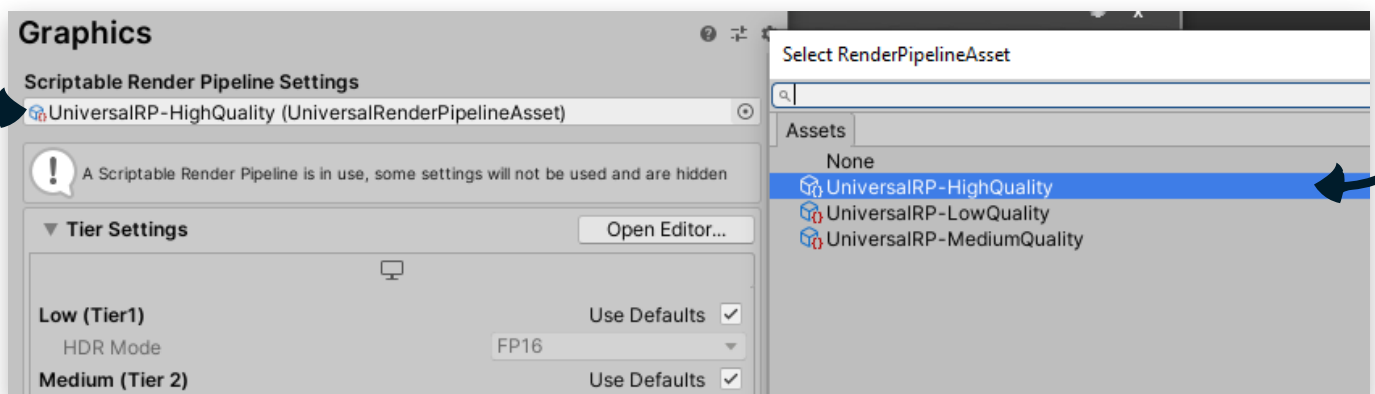
If you see any errors in the “Console”, try the “Clear” button. If the errors don’t disappear consult the FAQ or drop us an e-mail.

If you see any pink assets inside the Project window, simply select the said Prefabs (inside the prefabs folder) or the Meshes (inside the 3d folder) > right click > Reimport and it should fix it.

If you still encounter pink shaders, please make sure you have the correct pack version installed, depending on the render pipeline you are using.



Step 4: After the project is loaded, go to Edit > Project Settings...

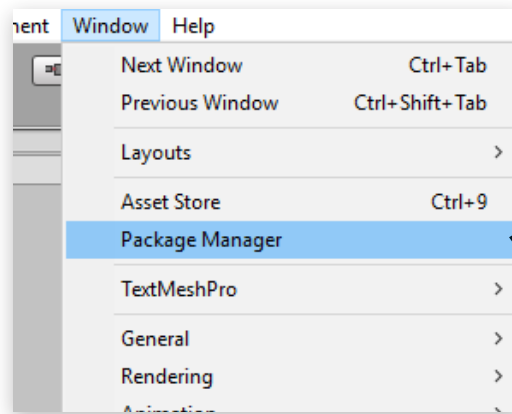


Step 5: For the Scriptable Render Pipeline Settings select “UniversalRP_HighQuality”. These are the presets Unity preinstalled with the Template.

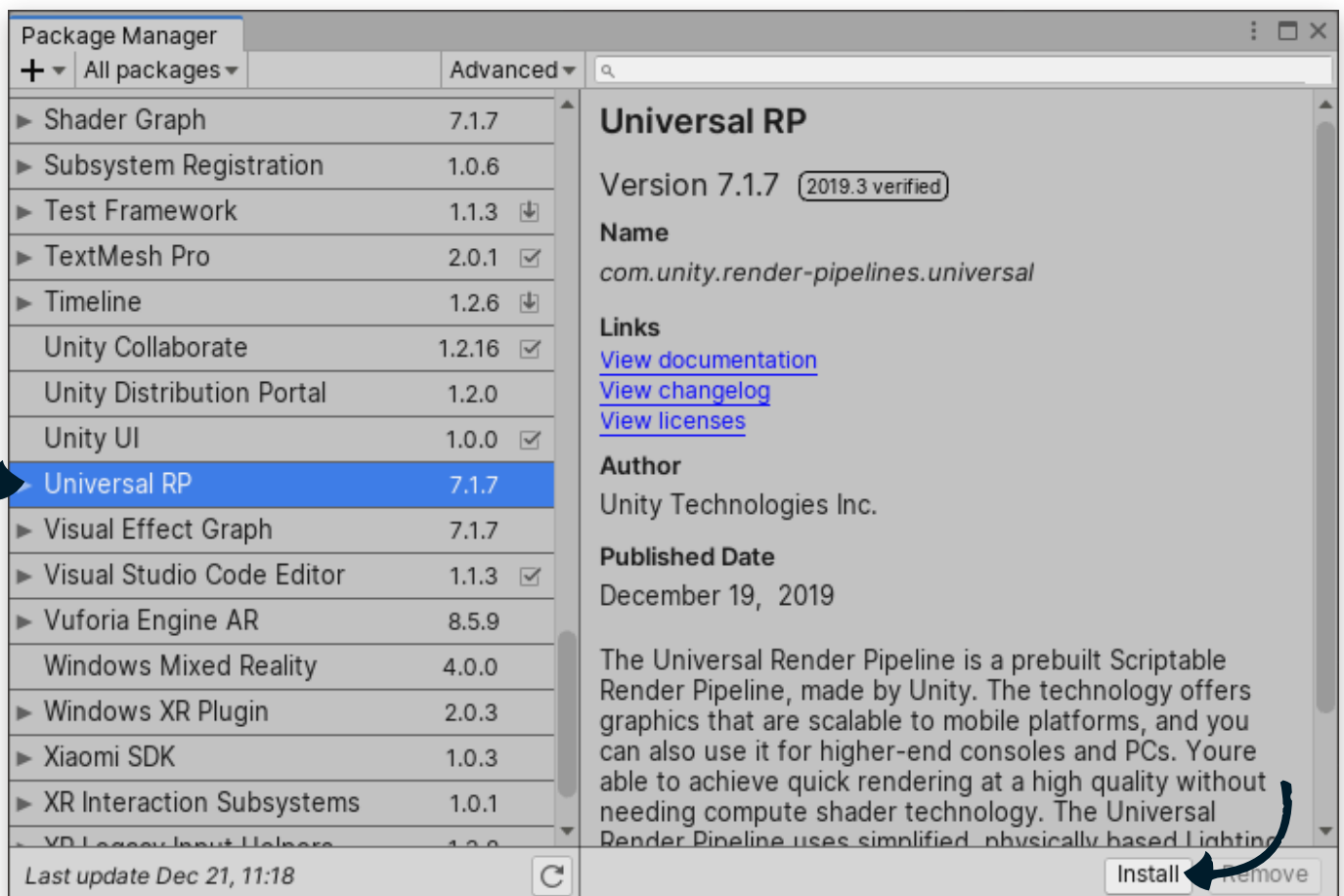


How to set up your project for URP (option 2)

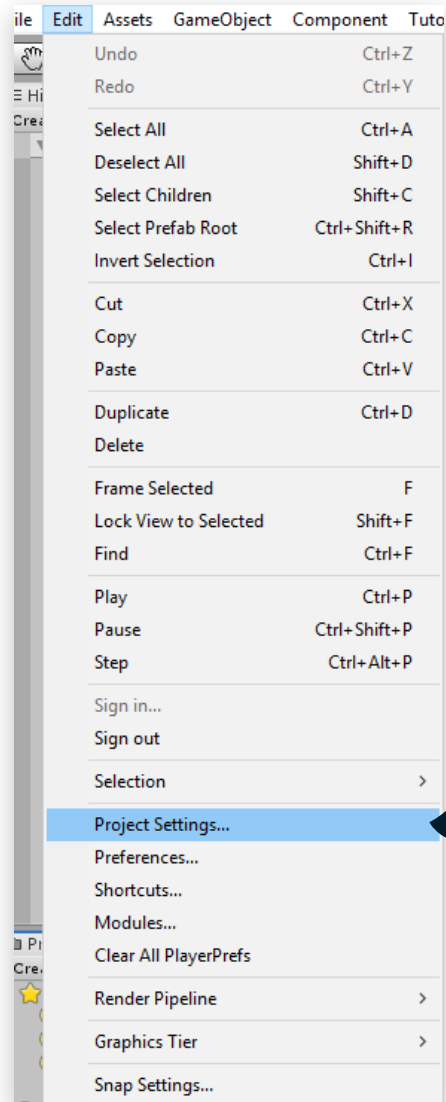
If you imported the pack before you installed the URP please follow the steps below:



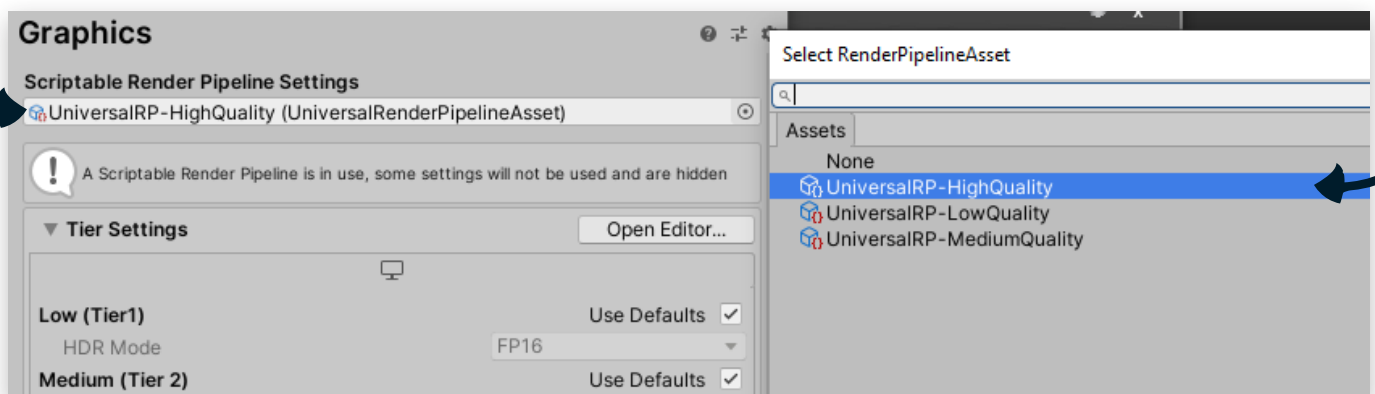
Step 1: go the Window > Package Manager.



Step 2: Select “Universal RP” asset and click “Install”.



Step 3: After the project is loaded, go to Edit > Project Settings...



Step 4: For the Scriptable Render Pipeline Settings select “UniversalRP_HighQuality”. These are the presets Unity preinstalled with the Template.





Demoscenes

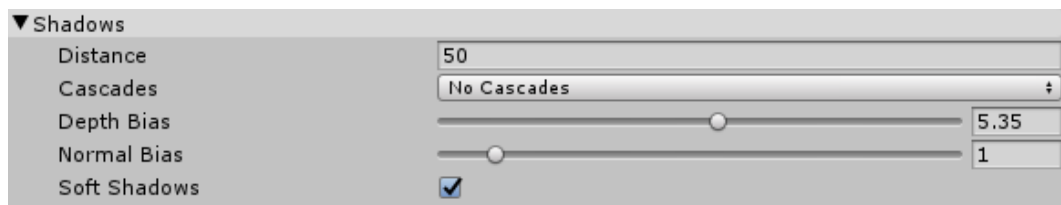
Demoscene_laserarena_level: Scene from the trailer and screenshots

Demoscene_laserarena_props: In this scene you will find all the assets within the package

Demoscene_laserarena_modular: In this scene you will find all the modular architecture assets within the package

Quality settings for URP

To quickly adjust any quality settings for URP please find the UniversalRP-HighQuality asset inside the /Assets/Settings folder.



Example settings for shadows in the render pipeline asset.

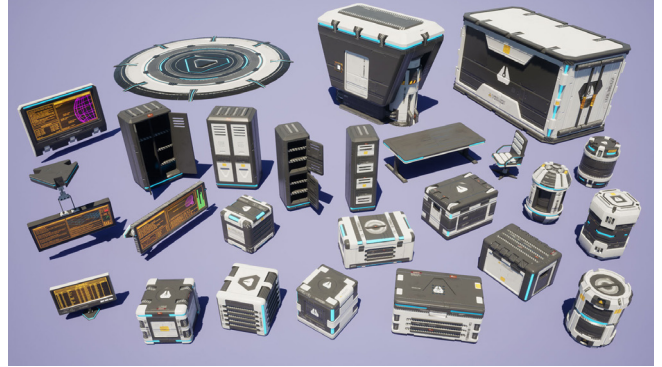
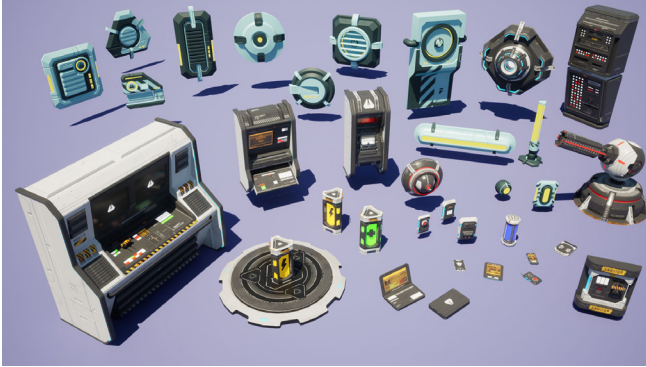
Post Processing

Inside the /Chroma Laser Arena/Settings folder you will find a Post Processing file for the demo scene. There you can adjust the postprocessing to your liking.



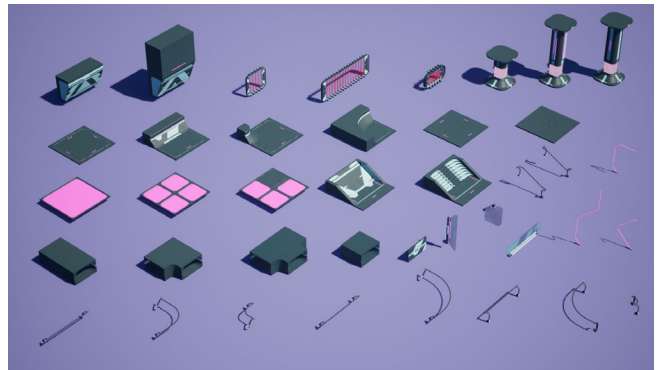
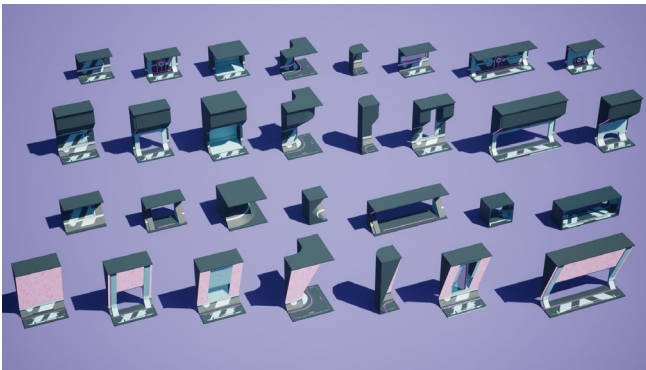
demoscene_laserarena_props

In this scene you will find all the props within this package.



demoscene_laserarena_modular

In this scene you will find all the modular assets within this package.

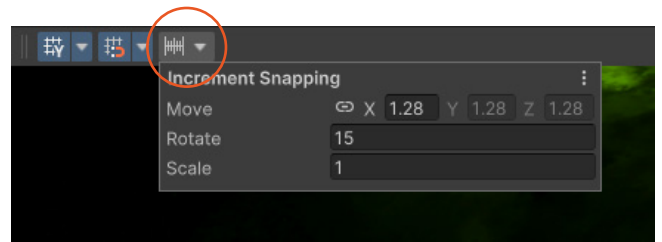
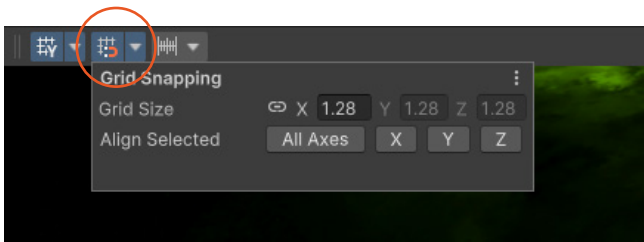




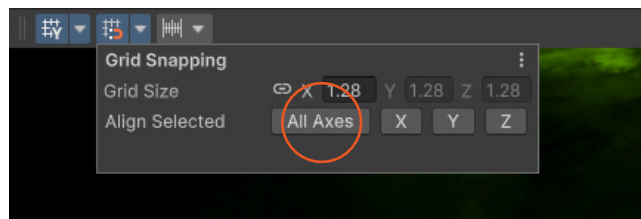
Grid snapping

IMPORTANT: All modular assets work with a Power of Two Grid.

For optimal grid snapping, make sure to adjust your grid snapping options. On the top left of your “Scene” Tab you will find the grid and increment snapping options. Any power of two value will work but for modular elements we suggest using the following values: 0.64, 1.28, 2.56, 5.12.



When placing modular assets in the scene, make sure that they are placed according to your grid snappings. The easiest way to do this is clicking on “All Axes” button in your grid snapping window.



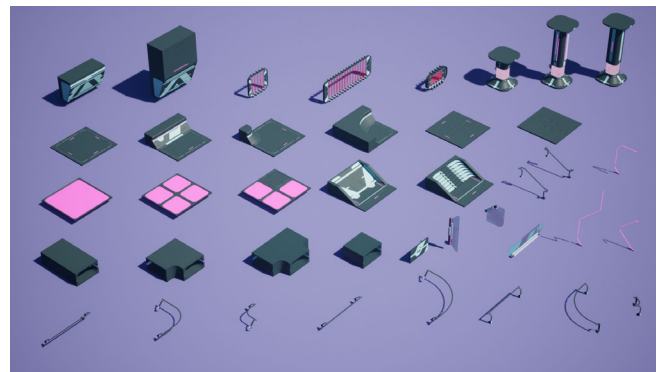
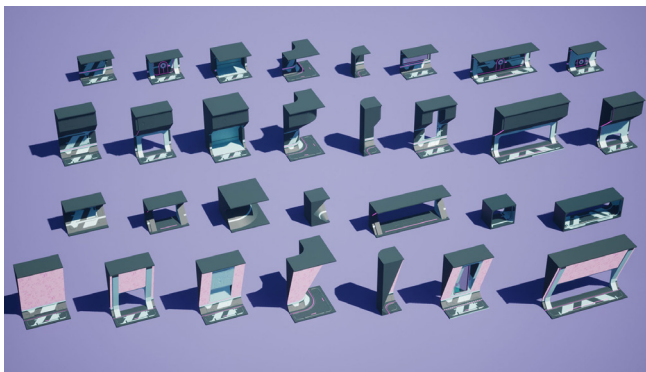


Naming convention

Prefixes and suffixes

All the modular pieces follow a strict naming convention to make your life easier. You can use the prefixes/affixes to either search for a group of assets, or simply know in the scene view which asset is what.

To familiarize yourself better with the naming convention we suggest looking into **demoscene_laserarena_modular**. There you will find every single piece of the pack:



Some assets correspond to wall variants by the number in the naming. For example

SM_MOD_Laser_Door_01

SM_MOD_Laser_DoorFrame_01

SM_MOD_Laser_Door_01_EnergyGate

SM_MOD_Laser_Door_01_EnergyGateLaser

are combineable with

SM_MOD_Laser_Wall_01_Door_01

These specific door assets and _02 variants of door assets are also used for Wall_04 variants.



Assets

Meshes

All props have 3 LOD levels and most of the modular assets have 4 LOD levels. All assets have either generated or custom mesh colliders and a custom Lightmap UV in the second channel. The Light Map Resolution for each mesh has been adjusted inside the mesh editor to best fit its mesh size.

UV Channel 2 (3rd channel) is used for Normal map details.

Textures & Materials

You can find all the textures in the /2d folder. The materials are in the /materials folder.

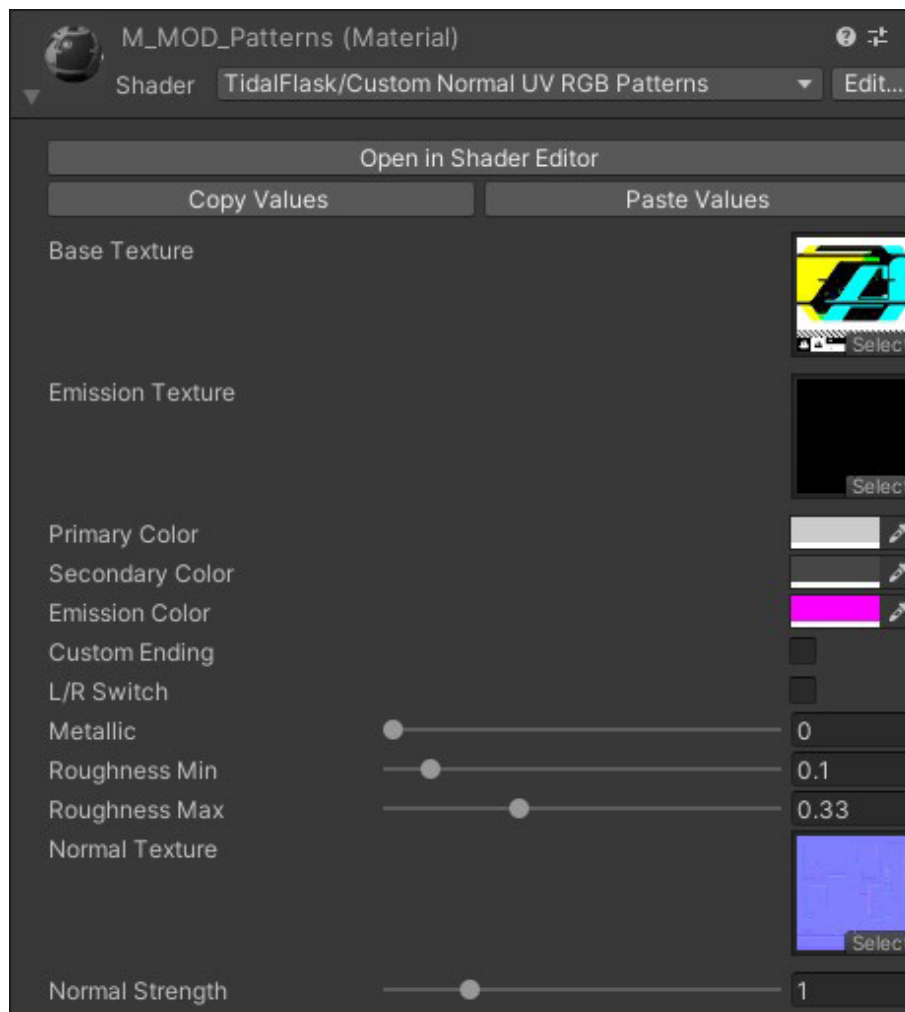
Some of the textures (T_MOD_Laser_Patterns and all _MASK textures) were created to be used with specific shaders with different information in each of their RGB channels. More about custom shaders below.



Physically Based Rendering (PBR)

For modular architecture assets there is no individual PBR map. Instead the _Patterns texture is used as a mask to differentiate between roughness values.

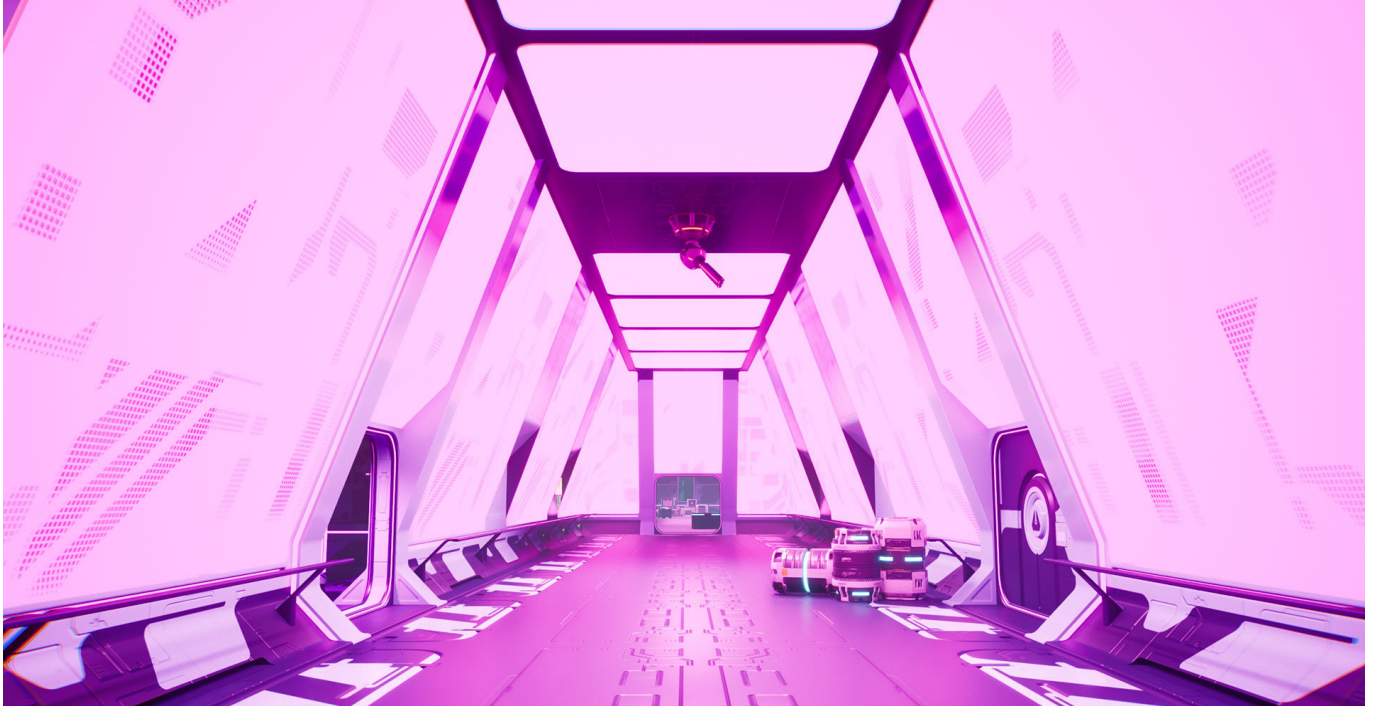
For props you have the controls for the min/max values of roughness. No asset has metalness per default but you can change the value in the material if needed.





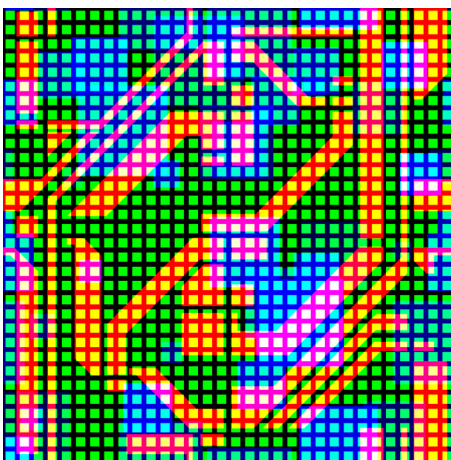
Animated emissive material

For Wall_04 variants of the architectural assets we have an animated material. The same shader is also used for the column's emissive segment.

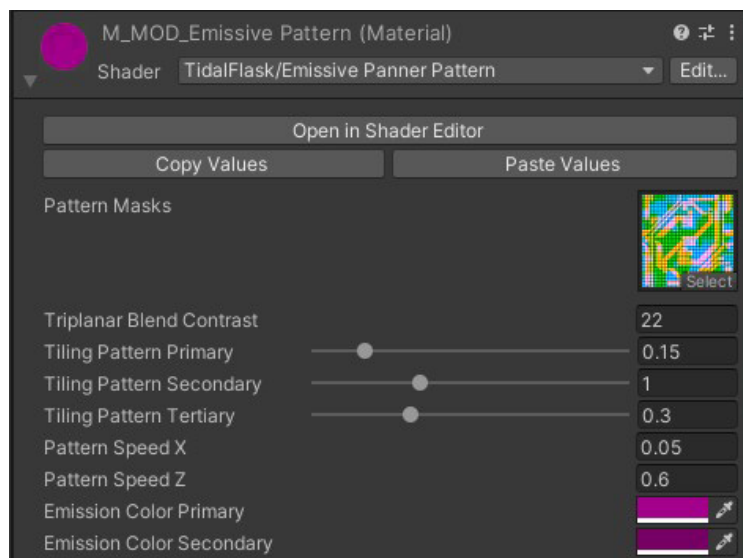


Animated emissive material for Wall_04 variant.

The material uses one single texture for the patterns with different RGB channels. RGB representing Primary/Secondary/Tertiary patterns. You can change color/tiling of the individual patterns and the speed of the tertiary mask to reveal the details.

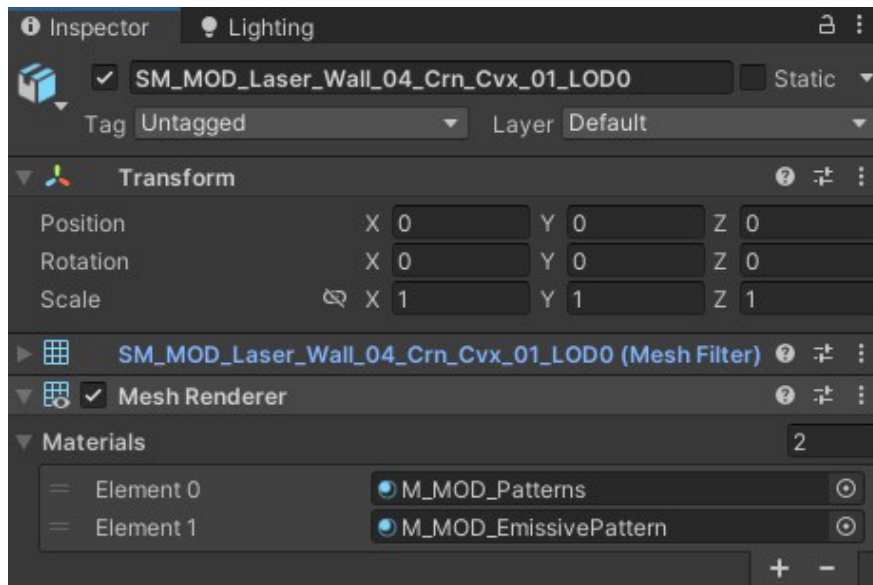


*R: Primary: main techy pattern
G: Secondary: square grid
B: Tertiary: animated mask to reveal details.*





If you don't want to use this material on the assets, you can add the same material into the 2nd slot (M_MOD_Patterns). The assets are prepared to only use the M_MOD_Patterns material by themselves. By using the same material, these surfaces will become fully emissive.





Shaders

The custom shaders were created using Amplify Shader Editor with Unity version 2021.3.20 (built-in) and 2021.3.2 (URP) and hence can **not** be opened or adjusted using Unity's Shader Graph.

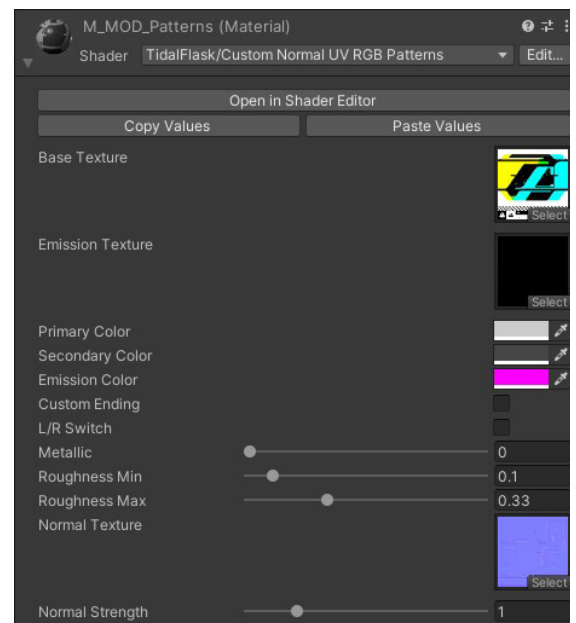
If you have Amplify Shader Editor installed, you can adjust the shaders there.

The rest of the shaders are all standard URP or Built-in, depending on which render pipeline you are using.

IMPORTANT: In case you are using the new shaders with a Unity version older than 2019.4.30 please be aware that this might result in shadow cascade errors in the scene. To solve the problem you can set the Cascades option in your render pipeline asset to "No Cascades".

Shader for modular assets:

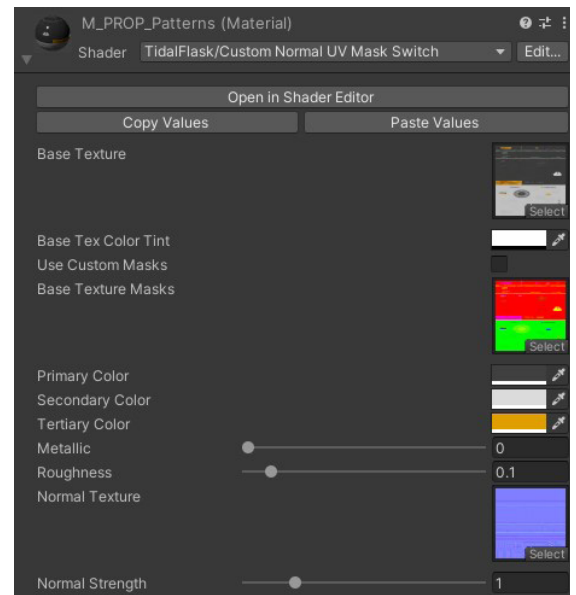
- Base Texture: slot for the base rgb mask texture
- Emission Texture: b/w emission mask
- Primary Color: primary color (base texture mask)
- Secondary Color: secondary color (base tex. mask)
- Emission Color: emission color (controlled by base tex. mask + emission texture)
- Custom Ending: switch to enable L/R channels in the base texture
- L/R Switch: switch between L/R endings of base texture
- Metallic: metallic value
- Roughness Min: minimum clamp for roughness (controlled by base texture mask)
- Roughness Max: maximum clamp for roughness (controlled by base texture mask)
- Normal Texture: slot for normal texture
- Normal Strength: increase/decrease normal strength





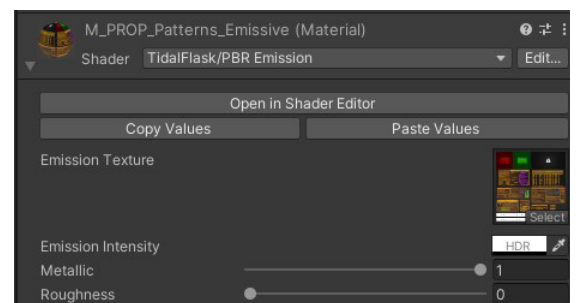
Shader for props:

- Base Texture: slot for base texture
- Base Tex Color Tint: tint color for base texture
- (only applicable if “Use Custom Masks” = false)
- Use Custom Masks: switch between base texture/ base texture masks use
- Primary Color: primary color (base texture mask) (only applicable if “Use Custom Masks” = false)
- Secondary Color: secondary color (base tex. mask) (only applicable if “Use Custom Masks” = false)
- Tertiary Color: tertiary color (base texture mask) (only applicable if “Use Custom Masks” = false)
- Metallic: metallic value
- Roughness: roughness value (not controlled by texture)
- Normal Texture: slot for normal texture
- Normal Strength: increase/decrease normal strength



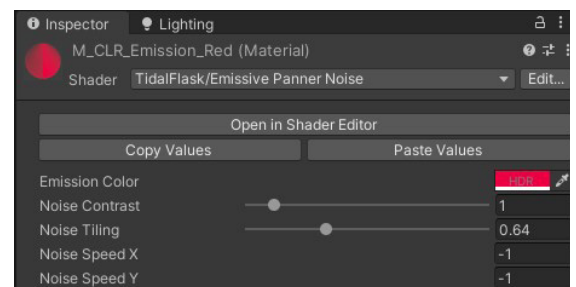
Shader for props:

- Emission Texture: slot for emission texture
- Emission Intensity: intensity controlled by color input
- Metallic: metallic value
- Roughness: roughness value



Shader for laser gate lasers:

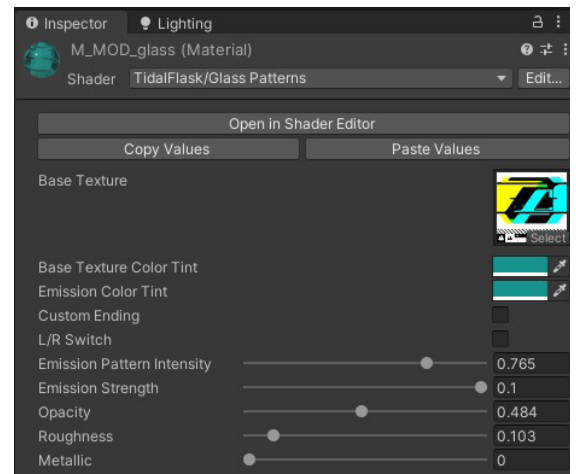
- Emission Color: emission color/intensity
- Noise Contrast: contrast control for animated noise
- Noise Tiling: tiling amount for animated noise
- Noise speed X: noise speed in U direction (mapped to UVs)
- Noise speed Y: noise speed in V direction (mapped to UVs)





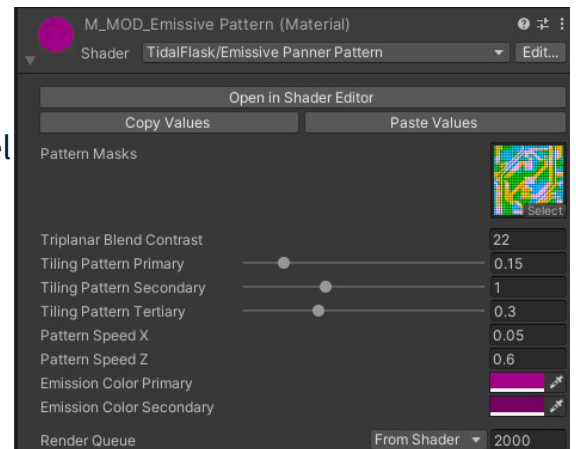
Shader for glass:

- Base Texture: slot for base texture
- Base Texture Color Tint: color tint for patterns
- Emission Color Tint: emission color for glass (not including patterns)
- Custom Ending: switch to enable L/R channels in the base texture
- L/R Switch: switch between L/R endings of base texture
- Emission Pattern Intensity: emission value for patterns
- Emission Strength: emission strength for glass (not including patterns)
- Opacity: glass opacity (including patterns)
- Roughness: roughness value (including patterns)
- Metallic: metallic values



Shader for animated emissive walls (and columns):

- Pattern Masks: RGB mask texture
- Triplanar Blend Contrast: contrast for X/Z blending
- Tiling Pattern Primary: tiling amount for R channel
- Tiling Pattern Secondary: tiling amount for G channel
- Tiling Pattern Tertiary: tiling amount for B channel
- Pattern Speed X: speed in X world axis
- Pattern Speed Z: speed in Z world axis
- Emission Color Primary: background color
- Emission Color Secondary: pattern color



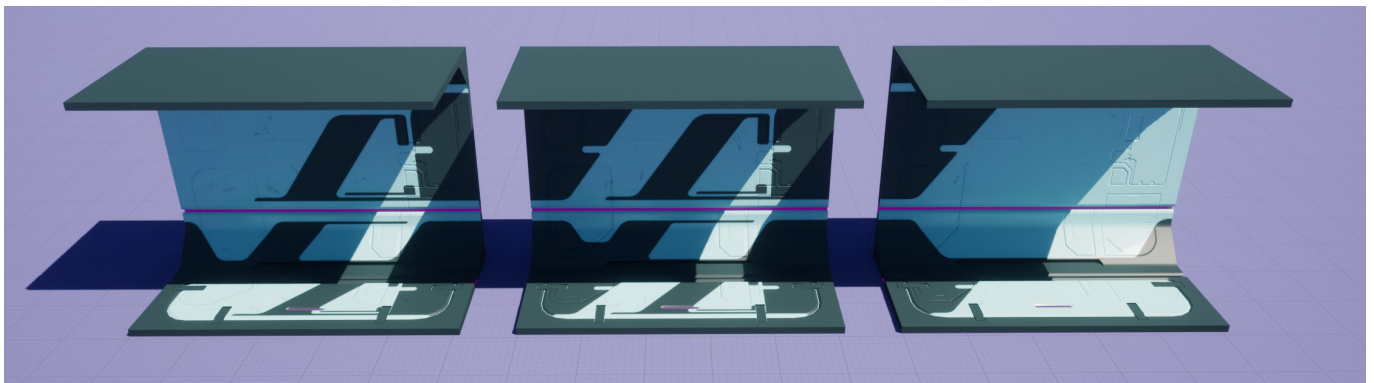


Customizing Modular Assets

For modular architecture pieces you can add additional variations by changing the main, secondary colors, emission color and you can chose a different pattern variation by changing the switches in the material instances. The channels in the `_Patterns` texture are as follows:

- R: more pattern on the right side
- G: default pattern, both sides are filled up
- B: more pattern on the left side

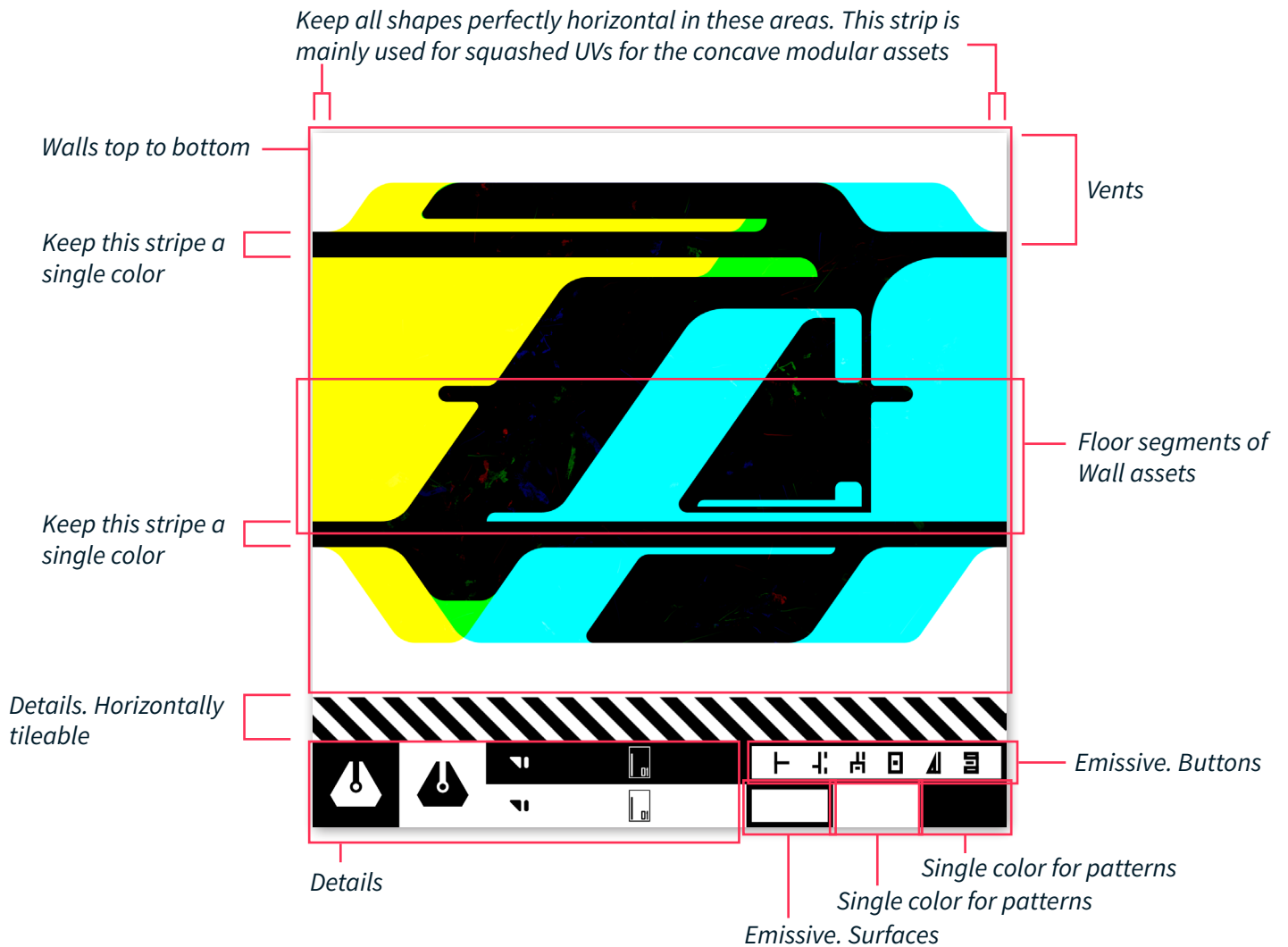
By toggling “Custom Ending” you activate the L/R versions. With the second switch called “L/R Switch” you can chose which ending you would like to have.



Custom ending and L/R variation. The pattern variation is visible on all modular pieces except `_Cve` wall assets, floors/ceilings which don't have a visible pattern.



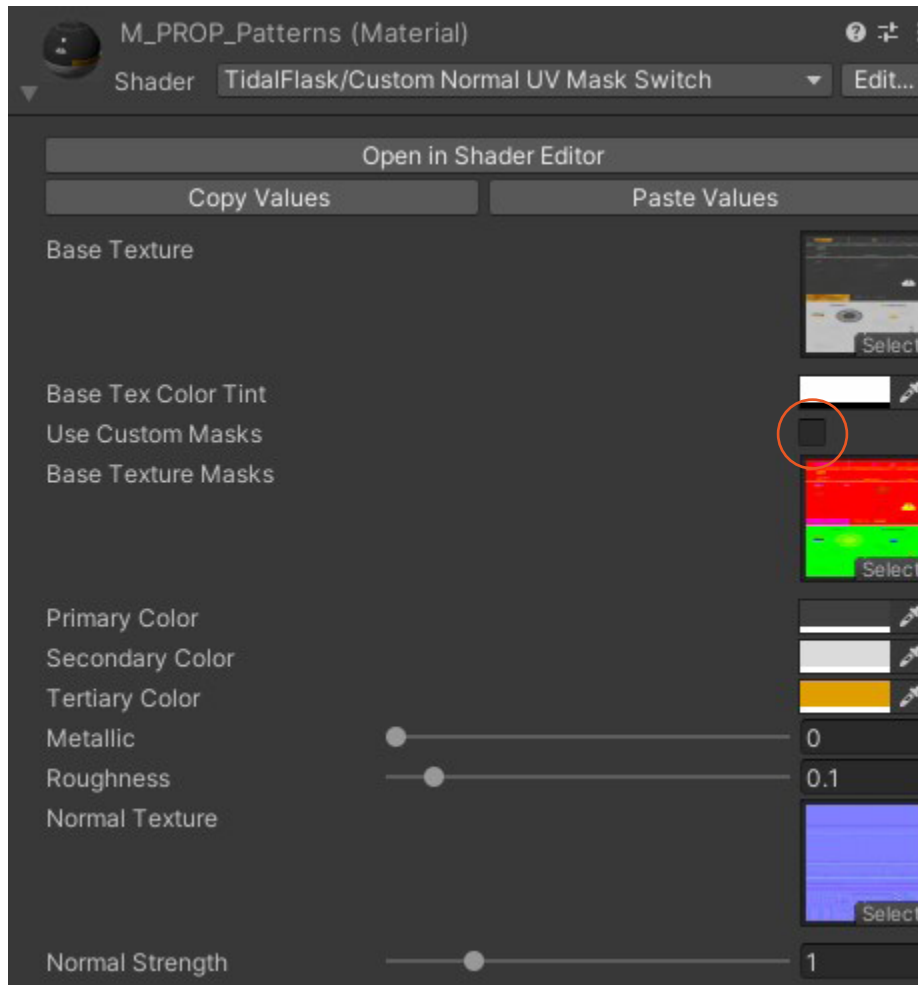
If you would like to create your own pattern please follow the layout below:





Customizing Props

Similarly to the modular assets, you can optionally change the colors of the props by enabling the “Use Custom Colors” toggle in the material instances. This way, instead of the base color texture the material uses an RGB mask to color the primary/secondary/tertiary colors:





Support

FAQ

Will there be updates to the package?

Yes. We plan to update all our packages as soon as there is a relevant update or if the community asks for adjustments.

Can you give support to users if something doesn't work?

Yes, but first please read through this document and if you still need help with something related to this package, feel free to contact us.

What's the deal with Universal Render Pipeline (URP)?

With Unity 2019.3 the Lightweight Render Pipeline is renamed to Universal Render Pipeline. If you set up your project using LWRP from an older version of our pack, you can change to URP and everything should work from the getgo - shaders, materials and lighting are compatible with URP.

A list of errors shows up in a shader.

Try reimporting the shader (in project tab > right-click on the shader > Reimport). We are aware of some shader warnings showing up, which don't seem to actually break the shader. So simply clearing the warning in the console tab should fix the problem.

I opened the project for the first time and everything is pink. When I select a material, the shader says "Hidden/InternalErrorShader"

This is the case when your project doesn't use the same render pipeline as the pack version you installed. Starting on page 4 you will find all the steps needed to properly set up your project.



I opened the project for the first time and in the Console I get the error "A tree couldn't be loaded because the prefab is missing"

This is a known Unity bug (importing a package that has terrain and trees in it) and has nothing to do with the package. Simply press "Clear" in the "Console" tab and it won't appear again.

I imported the package but some assets still appear pink in the scene...

Make sure you installed the correct render pipeline version of our pack. After opening a scene it's still possible, that some assets are pink. If that is the case, do the following:

- In the Hierarchy window select "Terrain"
- In the "Paint Details" tab double click on any asset
- Click on the circle next to the asset which was added in the "Detail" panel
- Re-add the same asset and the scene should look normal again

I imported the package but some assets still appear pink in the Project window...

If you see any pink assets inside the Project window or inside the "Terrain"-object in any of the scenes simply select the said Prefabs (inside the prefabs folder) or the Meshes (inside the 3d folder) > right click > Reimport and it should fix it.



Contact & Support

Visit our page for updates and more packages in the future:

<https://tidalflask.com/>

Contact us if you didn't find an answer to your questions:

info@tidalflask.com

Social Media



<https://www.facebook.com/tidalflask>



<https://twitter.com/TidalFlask>



<https://www.instagram.com/tidalflask/>

